

# AgentifAI Buildathon 2026

## Slide 1: Team Composition

Team Name:

**Smooth  
Operators**

Problem Statement:

**#02 Personal Research Assistant**

**Shriansh Vikram Singh**

Team Lead & AI Architect | B.Tech CSE, 3rd Year

**Arunaabh Mojtra**

Backend & RAG Engineer | B.Tech CSE, 3rd Year

**Shivansh Gupta**

Multi-Agent Engineer | B.Tech CSE, 3rd Year

**Saloni Singh**

Research Lead | B.Tech CSE, 3rd Year

**Mahi Chauhan**

Frontend & UX Engineer | B.Tech CSE, 3rd Year

# The **\$8.5B** Problem Nobody Has Solved Yet

## The Research Overload Crisis

**60%** of research time is wasted organizing information rather than generating insights.

- » **Information Explosion:** 2.5 million research papers published yearly create unmanageable data chaos.
- » **Validation Bottleneck:** Researchers constantly face contradictory findings across fragmented, unverified sources.
- » **Tool Failure:** ChatGPT hallucinates citations. Perplexity provides shallow synthesis. Google Scholar is merely retrieval.

⚠ This is not a search problem. This is a reasoning problem.



# SynaptiQ

## ResearchOS

### Autonomous Multi-Agent Pipeline

Sequential reasoning agents collaborate to generate grounded research intelligence.

### Claim-Level Verification

Every insight is validated using citations and contextual grounding.

### Research Gap Detection

Identifies unexplored opportunities and contradictory findings.

### Enterprise AI Stack

GPT-4, LangChain, Azure OpenAI, FAISS, FastAPI, Next.js

## Multi-Agent Flow

User Query



Discovery Agent



Verification Agent



Comparative Analysis



Executive Research Brief

# Innovation Story:



## The Synthesis Bottleneck

Researchers drown in a sea of PDFs, wasting 70% of their time on manual organization. We identified that the problem isn't lack of information, but the lack of Research Intelligence.



## Beyond Static RAG

SynaptiQ evolves beyond simple search. We built an Autonomous Multi-Agent Pipeline where discovery, verification, and analysis agents collaborate to simulate human reasoning at scale.



## Mapping the Unseen

Our solution doesn't just summarize; it detects research gaps and contradictory findings, transforming raw data into a visual literature graph for instant breakthrough detection.



## EXECUTION APPROACH

# Built to Scale

### Phase 1 — Core RAG

Document ingestion, vector indexing, semantic retrieval.

### Phase 2 — Orchestration

Multi-agent workflows and contextual routing.

### Phase 3 — Intelligence Layer

Contradiction detection and synthesis pipeline.

### Phase 4 — Deployment

Cloud-native Azure deployment with observability.

## Success Metrics

92%+ Accuracy

95% Citation Precision

<8s Latency

80% Hallucination Reduction

OpenTelemetry Monitoring

Azure Cloud Deployment